Traffic Collision Data Analysis

1. **Introduction**
   This project analyses California traffic collision data using Apache Spark and PySpark to process large-scale datasets efficiently. The objective is to clean, transform, and explore collision records to identify accident patterns, severity trends, and contributing factors. MinIO (instead of Amazon S3) object storage is used to store processed data for scalable access, backup, and future analysis. The insights generated can support traffic safety improvements, urban planning, and policy decision-making.

2. **Objective**
   The objective of this project is to analyse California traffic collision data using Apache Spark and PySpark for large-scale data processing. The project focuses on cleaning, transforming, and exploring collision records to identify accident patterns, severity trends, and contributing factors. MinIO object storage is used to store processed datasets after ETL operations for scalable and efficient access. The final goal is to generate actionable insights that can support traffic safety planning and better decision-making.

3. **Business Value**
   Traffic collisions create major risks to public safety, infrastructure, and transportation efficiency. By analyzing historical collision data, authorities can identify high-risk locations, dangerous time periods, and common causes of accidents. These insights help government agencies and urban planners improve road design, optimize traffic management, and enhance emergency response planning. Data-driven strategies can ultimately reduce accident frequency, improve commuter safety, and support smarter city planning.

4. **Dataset Overview**
   The dataset contains California traffic collision records collected from the Statewide Integrated Traffic Records System (SWITRS). It includes detailed information about accidents such as severity, involved vehicles, victims, environmental conditions, and locations. The dataset was originally stored in SQLite format, then a sample of the data is taken for further processing using PySpark for scalable analytics. It is divided into multiple relational tables for structured analysis.

4.1 Data Source

The data used in this project is sourced from the California Statewide Integrated Traffic Records System (SWITRS). The original dataset is provided in SQLite database format containing historical traffic collision records across California. It includes structured tables with information related to accidents, involved parties, victims, and locations.

4.2 Dataset Structure

The dataset is organized as a relational database containing multiple interconnected tables. Each table stores a specific category of collision data and can be linked using common keys such as case identifiers. This structure enables efficient querying, filtering, and joining of records for comprehensive analysis using Apache Spark.

4.3 Tables Description

- Collisions Table: The collisions table contains core information about each traffic accident, including date, time, severity, weather, lighting conditions, road surface, and number of victims involved.

- Parties Table: The parties table stores details about the people or vehicles involved in collisions, including age, gender, sobriety status, vehicle type, and at-fault indicators.
- Victims Table: The victims table contains injury-related information for individuals affected by collisions, including victim role, age, gender, injury severity, seating position, and safety equipment used.
- Locations Table: The locations table contains geographical and road-related details such as county, jurisdiction, road names, coordinates, and intersection information for each collision site.

5. Assignment Tasks

5.1 Data Preparation

To prepare the traffic collision dataset for analysis, I first installed the required libraries including PySpark and Pandas for large-scale distributed data processing. Since the files were stored in a MinIO object storage bucket, I configured a Spark session with the required Hadoop AWS and S3A connectors. This allowed Spark to securely connect to the MinIO bucket and directly read the CSV files.

The four structured tables loaded into Spark DataFrames were:

case_ids – unique case identifiers with database year

collisions – collision-level information such as severity, date, weather, road conditions, injuries, and location

parties – details about involved parties such as drivers, pedestrians, and vehicles

victims – victim-level records including injury severity, age, and seating position

All datasets were loaded using schema inference and header recognition to automatically detect column names and initial data types.

1. Schema Validation

I used printSchema() on all DataFrames to inspect column names and data types. This helped identify incorrectly inferred types such as:

- case_id stored as **double** instead of integer/long
- age fields stored as **double** instead of integer
- date and time columns requiring conversion into proper date/timestamp formats

2. Sample Record Inspection

Using .show(5), I reviewed sample records from each table to verify that:

- values were loaded correctly
- categorical columns contained readable labels
- no column shifting or parsing issues existed
- missing values were represented as NULL

3. Missing Value Analysis

A reusable function was created to calculate missing values for every column using:

- NULL checks
- blank string checks

This was performed for all four datasets. Then missing percentages were calculated using total row counts.

Examples of findings:

- latitude and longitude had over **71% missing values**
- reporting_district had over **59% missing values**
- some safety equipment columns had moderate missing values

5.2 Data Cleaning

1. **Fixing Columns**

   Columns with more than **50% missing data** were considered low quality and removed from further analysis.

   Examples of dropped columns from collisions table:

- latitude
- longitude
- reporting_district
- caltrans_county
- state_route
- postmile
- pcf_violation_subsection

   This improved data quality and reduced unnecessary dimensionality.

   After cleaning, columns were converted into appropriate formats:

- case_id → Long
- age fields → Integer
- injury counts → Integer
- collision_date → Date
- collision_time → Timestamp
- process_date → Date

  This ensured consistency for joins, aggregations, filtering, and time-series analysis.

2. **Handling Missing Values**

   After correcting schemas and removing sparse columns, the next step was to handle the remaining missing values so the dataset could be used reliably for analysis and modeling.

   2.1. Identifying Missing Values

   A custom PySpark function was used to calculate the percentage of missing values in every column across all cleaned tables:

- victims_clean
- parties_clean
- collisions_clean

  Missing values were defined as:

- NULL
- empty strings ("")

  This helped identify columns that still required imputation after initial cleaning.

  2.2. Numerical Value Imputation

  For numerical columns, statistical replacement methods were used.

  Mean Imputation

  Average values were calculated and used for age-related fields:

- victim_age → replaced with average victim age
- party_age → replaced with average party age

  This preserved the central tendency of the data without creating unrealistic values.

  Median Imputation

  For vehicle_year, the median was used instead of zero or mean because vehicle years are skewed and sensitive to extreme values.

- vehicle_year → replaced with median model year

  2.3. Categorical Value Imputation

  Missing categorical values were replaced with "Unknown" to preserve rows while clearly indicating unavailable information.

  Examples:

  Victims Table

- victim_sex
- victim_seating_position
- victim_safety_equipment_1
- victim_safety_equipment_2

- victim_ejected

Parties Table

- party_sex
- party_sobriety
- party_type
- direction_of_travel
- financial_responsibility
- vehicle_make
- party_race

Collisions Table

- chp_shift
- population
- county_location
- weather_1
- collision_severity
- type_of_collision
- lighting
- control_device

2.4. Numeric Sentinel Values

Some coded numeric fields were filled using logical default values:

- cellphone_in_use → 0
- jurisdiction → 0
- special_condition → 0
- killed_victims → 0
- injured_victims → 0

For some unknown coded values:

- intersection → -1
- tow_away → -1
- state_highway_indicator → -1

This preserved structure while distinguishing unknown values from valid zero values.

2.5. Time Field Imputation

Missing collision times were replaced with:

- 00:00:00

This maintained timestamp consistency for time-based analysis.

2.6. Final Validation

After imputation, missing-value percentages were recalculated. Results showed nearly 0% missing values across all cleaned datasets, confirming that the tables were ready for further analysis.

Outcome

Handling missing values improved:

- completeness of the dataset
- model readiness
- consistency in aggregations and joins
- reduced risk of errors during analysis

3. Handling Outliers

Once missing values were addressed, outlier detection and treatment were performed to reduce the impact of extreme or unrealistic observations.

3.1 Duplicate Removal Before Outlier Analysis

Duplicate records were first removed using primary keys.

Deduplication Keys Used:

- victims → id
- parties → id
- collisions → case_id
- case_ids → case_id

Duplicates Removed:

Table	Duplicates Removed
Victims	46,881
Parties	85,098
Collisions	20,577
Case_IDs	20,973

This ensured that repeated records did not distort outlier calculations.

3.2 Outlier Detection Method

Outliers were detected using the **Interquartile Range (IQR) method**.

Formula:

- IQR = Q3 - Q1
- Lower Bound = Q1 - 1.5 × IQR
- Upper Bound = Q3 + 1.5 × IQR

Spark's approxQuantile() was used for efficient percentile estimation on large datasets.

3.3 Columns Tested for Outliers

Victims

- victim_age

Parties

- party_age
- vehicle_year
- party_number_killed

- party_number_injured

Collisions

- distance
- injured_victims
- party_count
- injury count variables
- pedestrian / bicyclist / motorcyclist counts

3.4 Detection Results

Examples:

- victim_age had **2.27% outliers**
- party_age had **2.84% outliers**
- vehicle_year had **9.29% outliers**
- distance had **15.79% outliers**

Some variables were highly skewed with IQR = 0, so they were skipped.

3.5 Prudent Outlier Treatment

Instead of aggressively removing all extreme values, a balanced strategy was applied.

Standard Treatment (1.5 × IQR)

Used for normal distributions such as:

- victim_age
- party_age

Relaxed Treatment (3.0 × IQR)

Used for naturally variable columns to preserve data:

- vehicle_year
- distance
- injured_victims

Logical Lower Bounds Applied

Negative values were not allowed for count or age variables, so lower bounds were adjusted to:

- minimum = 0

3.6 Rows Removed After Cleaning

Table	Rows Removed
Victims	20,791
Parties	60,163
Collisions	96,171

Final Dataset Sizes

Table	Final Rows
Victims	896,261
Parties	1,721,656
Collisions	819,043

Outcome

Outlier handling improved data quality by:

- removing unrealistic ages and values
- reducing distortion in averages and trends
- preserving valid extreme real-world observations through prudent thresholds

The cleaned datasets are now statistically robust and ready for deeper exploratory analysis and machine learning.

5.3 Exploratory Data Analysis

1. Variable Classification

The variables were classified into:

Numerical Variables

- victim_age
- party_age
- vehicle_year
- killed_victims
- injured_victims
- party_count

- severe_injury_count
- pedestrian_killed_count
- bicyclist_injured_count
  etc.

  Categorical Variables
- collision_severity
- weather_1
- lighting
- road_condition_1
- party_type
- victim_role
- victim_sex
- county_location
  etc.

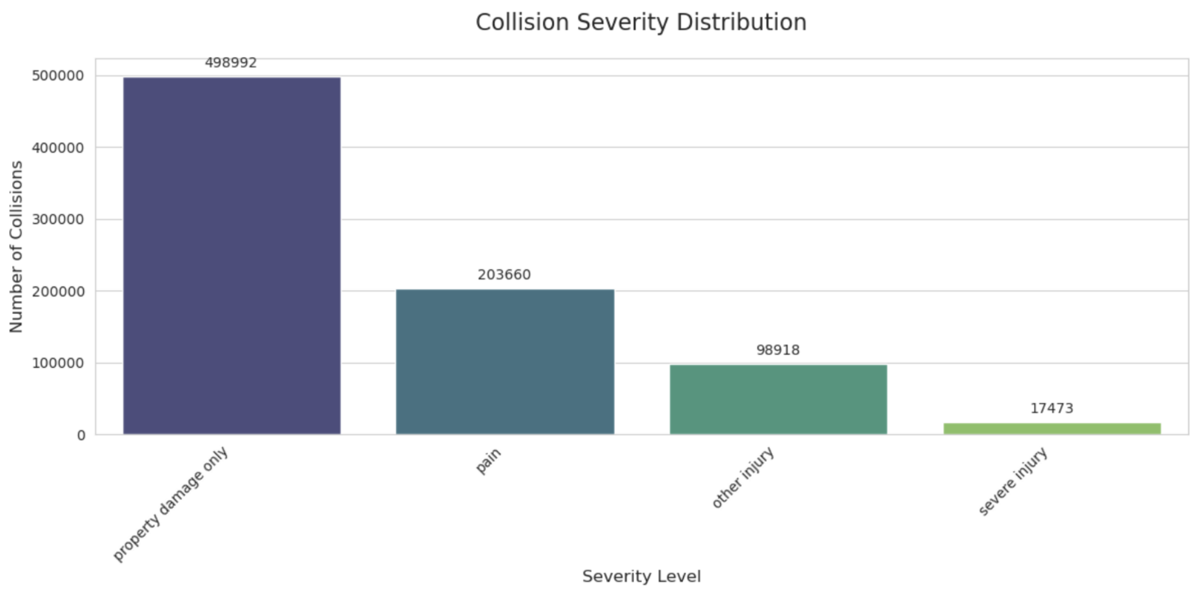
  Method Used

  Categorical variables were converted into numerical indexed values using **PySpark StringIndexer**.

  Final Output Tables
- victims_final
- parties_final
- collisions_final
- case_ids_final

2. Collision Severity Distribution

   A bar chart was created to visualize collision severity frequencies.



Analysis

The chart provides a macro-view of the outcomes of traffic accidents in California, categorized by the level of harm or damage caused.

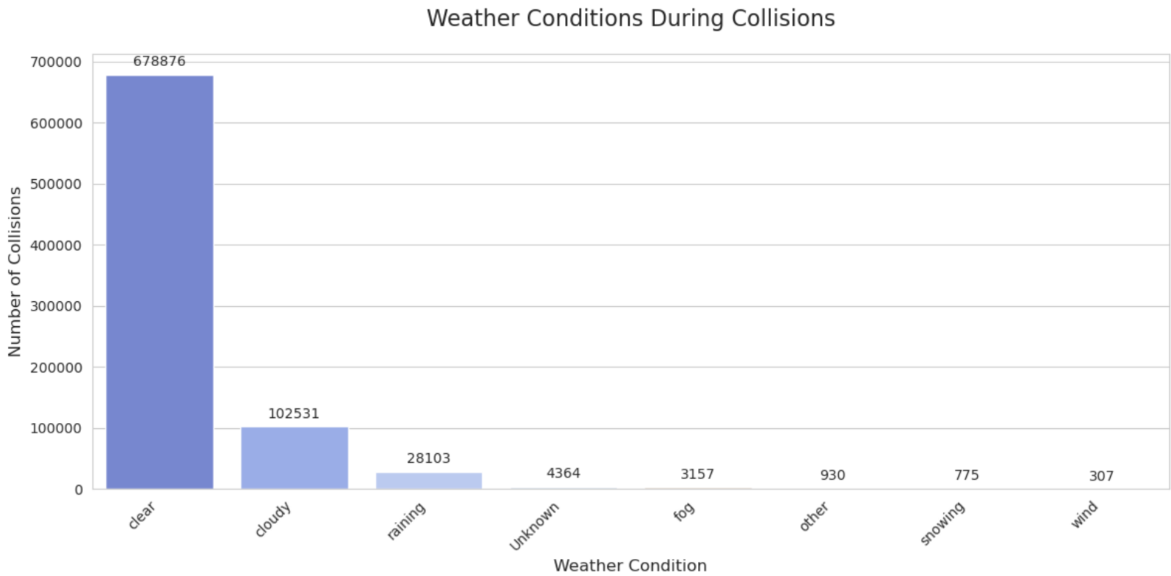
Prevalence of Non-Injury Collisions: Nearly 61% of the total collisions shown in this visualization resulted only in property damage. This indicates that while accidents are frequent, the majority do not result in immediate physical harm.

The "Injury Gap": There is a significant drop-off between "pain" and "severe injury." You are roughly 11 times more likely to be involved in a collision involving "pain" than one involving a "severe injury."

Safety Implications: The relatively low number of "severe injuries" compared to the total volume suggests that safety features (airbags, crumple zones) and lower-speed urban environments may be effectively mitigating the most extreme physical outcomes, even though the overall accident volume remains high.

3. Weather Conditions Analysis

Collision counts were grouped by weather condition.



Analysis

The "Clear Weather" Dominance: A staggering 678,876 collisions occurred under clear skies. This suggests that weather is not the primary driver of accidents; rather, human factors (distraction, speed, volume) play a much larger role.

Adverse Weather Minimal Impact: Combined, accidents during rain, fog, snow, and wind make up a tiny fraction of the total. For example, there are roughly 24 times more accidents in clear weather than in the rain.

The Exposure Bias: It is important to note that these numbers likely reflect exposure rather than risk level. It is clear more often than it is raining in California, so more "clear weather" miles are driven, leading to a higher raw count of accidents.

4. Victim Age Distribution

   Histogram plotted for victim ages.

Analysis

The "Young Driver" Peak (Ages 18–25): The highest frequency of victims occurs in the late teens and early twenties, specifically peaking around age 19–21. This aligns with insurance industry data regarding high-risk categories: newer drivers with less experience and a higher likelihood of being on the road during late-night hours.

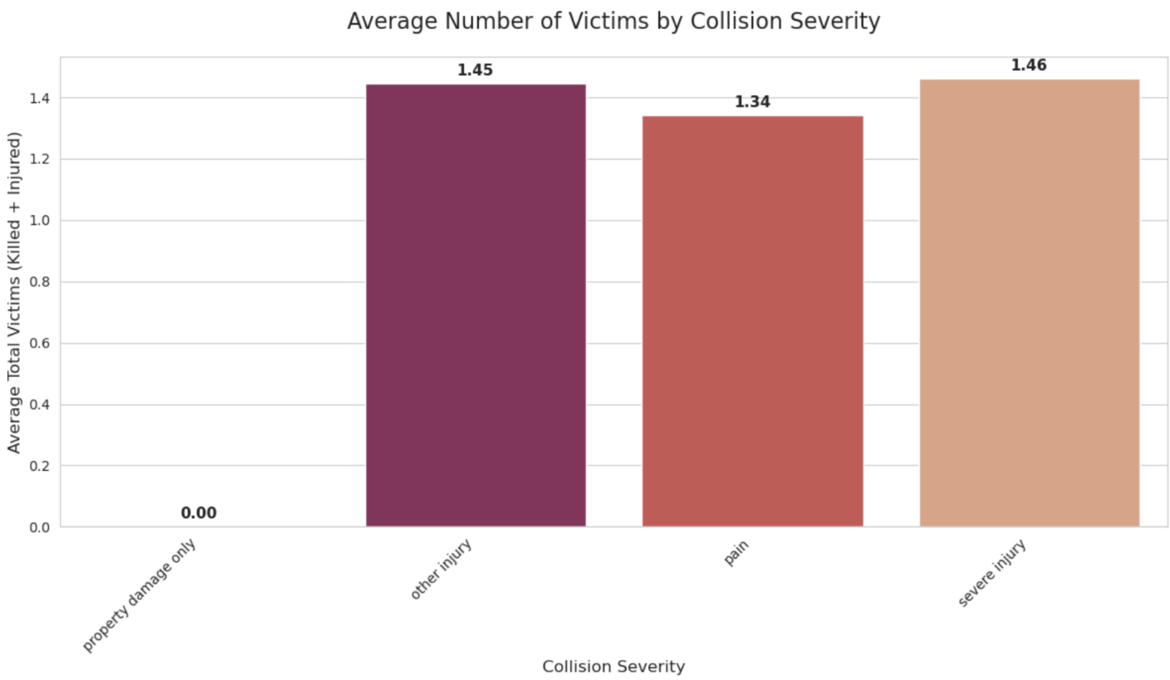
The "Age 30" Spike: There is a very sharp, isolated spike right at age 30. In large datasets like SWITRS, a localized spike like this might indicate "data rounding." When an officer or victim is unsure of an exact age, they may default to a round number like 30.

The Child/Adolescent Dip: There is a noticeable dip in victims between the ages of 13 and 15. This reflects the transition period where children are no longer as frequently passengers in family vehicles but are not yet old enough to drive themselves.

Steady Decline Post-35: After age 35, there is a consistent, gradual decline in the number of victims. This likely reflects a combination of increased driving experience, changes in risk-taking behaviour, and a general reduction in total miles driven as people age.

5. Severity vs Number of Victims

Average number of victims was compared across severity categories.



Property Damage Only (0.00): This is a logical baseline. By definition, a "Property Damage Only" collision involves zero injuries or fatalities.

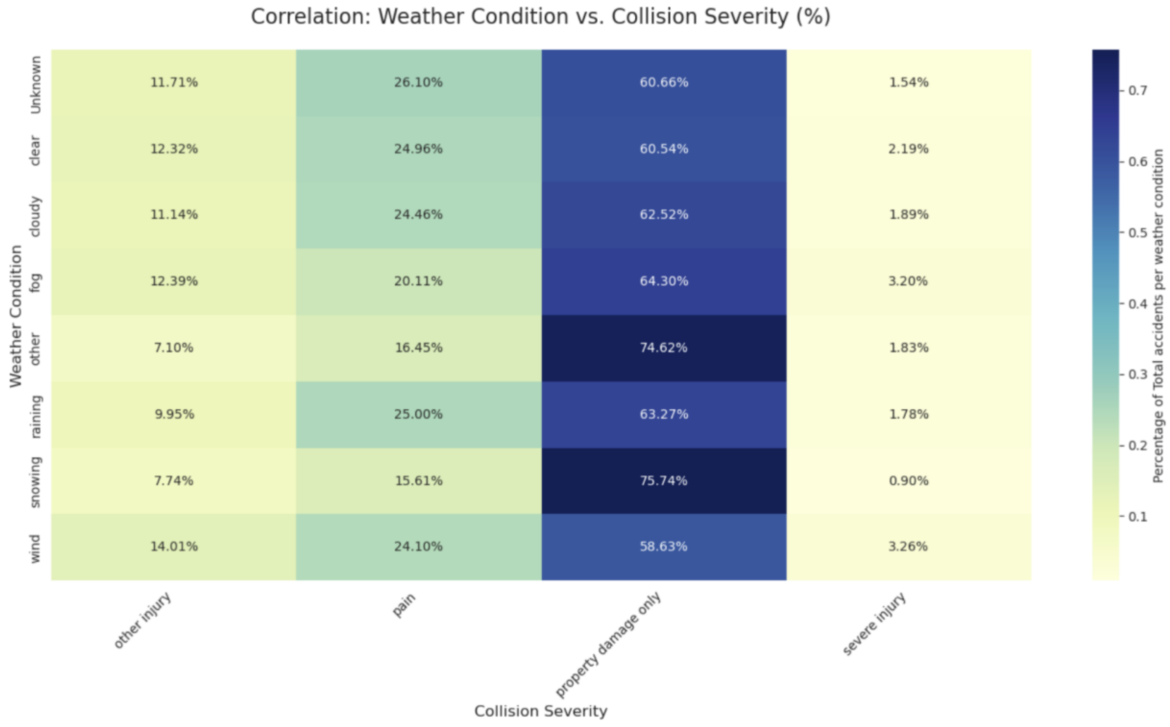
Severe Injury (1.46): This category has the highest average. This suggests that high-severity collisions often involve multiple people (e.g., a driver and a passenger, or a multi-vehicle pile-up).

Other Injury (1.45): Interestingly, the average number of victims for "other injury" is nearly identical to "severe injury." This indicates that the number of people affected doesn't necessarily dictate the severity of the physical outcome; a single event often results in multiple people sustaining moderate injuries.

Pain (1.34): This has the lowest average among the injury categories. It suggests that "pain-only" reports might more frequently involve a single occupant or fewer people per incident compared to more violent crashes.

6. Weather vs Severity Correlation

   Heatmap used to compare severity percentages under each weather condition.



The "Property Damage" Baseline: Across almost all weather conditions, "Property Damage Only" remains the dominant outcome, hovering around 60% to 75%. This reinforces that the majority of traffic incidents, regardless of weather, are non-injurious.

The Snowfall Paradox: Interestingly, Snowing shows the highest percentage of "Property Damage Only" at 75.74%, and the lowest percentage of "Severe Injury" at 0.90%.

Interpretation: Drivers likely slow down significantly during snow, leading to more "fender benders" but fewer high-speed, high-impact severe injuries.

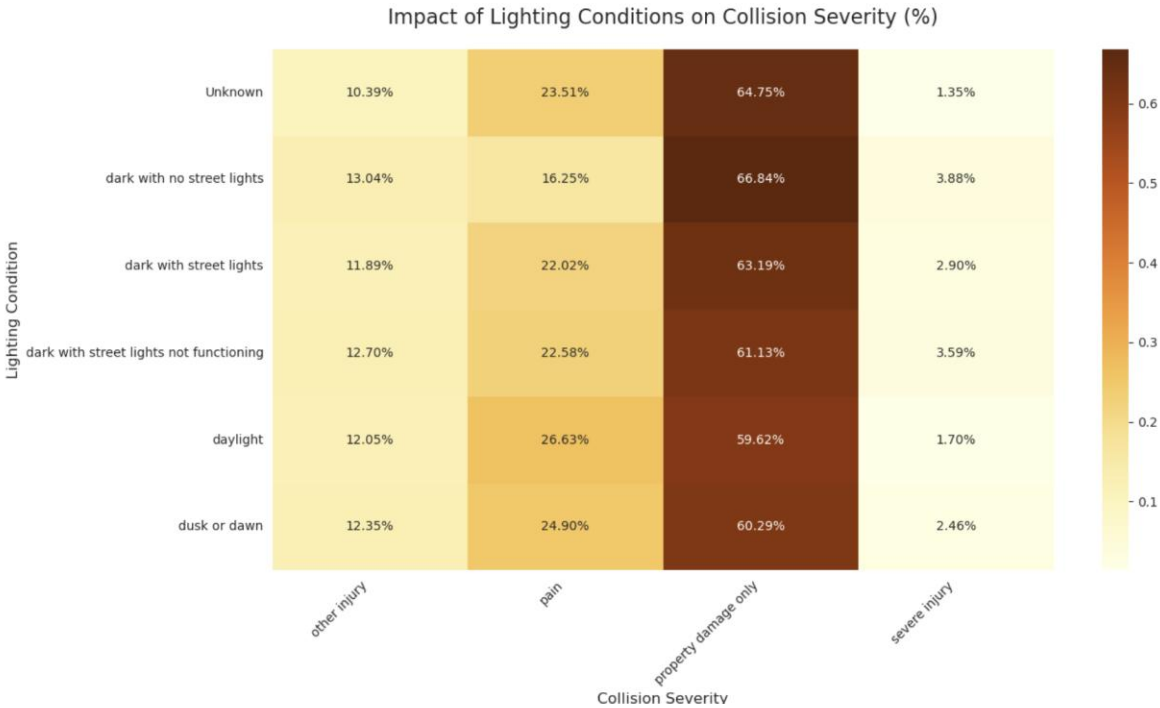
Fog and Wind Risks: Wind shows a spike in "Severe Injury" (3.26%), the highest on the chart. High winds can cause vehicle instability or rollovers, especially for high-profile vehicles.

Fog shows a higher rate of "Other Injury" (12.39%) and "Severe Injury" (3.20%) compared to clear weather. Reduced visibility often leads to higher-speed rear-end collisions.

Clear vs. Raining: The severity distribution for Clear and Raining is remarkably similar. This suggests that while rain increases the probability of a crash occurring (slippery roads), it doesn't necessarily change the severity profile of the crash compared to a clear day.

7. Lighting Impact on Severity

Lighting conditions compared against collision severity.



The Danger of Total Darkness: The category "dark with no street lights" shows the highest rate of Severe Injury (3.88%). When drivers lack artificial illumination, reaction times are likely slower, and the impact speeds are higher, leading to more critical outcomes.

The Effectiveness of Infrastructure: Comparing "dark with street lights" (2.90% severe) to "dark with no street lights" (3.88% severe) shows a measurable safety benefit to road lighting. Interestingly, when street lights are present but not functioning, the severe injury rate jumps back up to 3.59%.

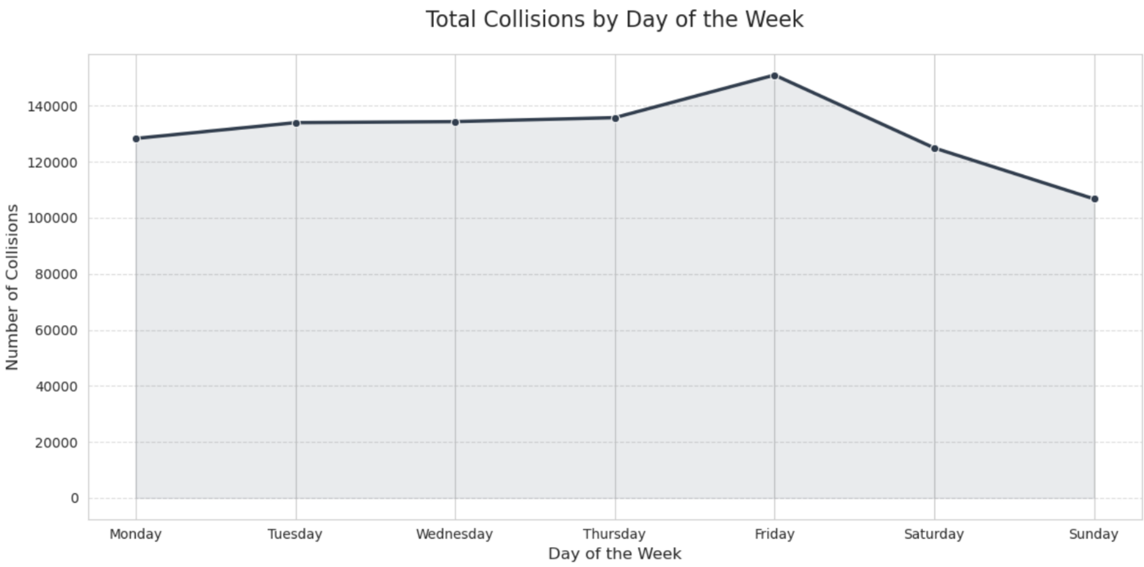
Daylight Paradox: While daylight has the highest volume of total accidents (as seen in typical traffic patterns), it actually has one of the lowest severe injury rates at 1.70%.

Insight: Congestion during daylight hours might lead to more frequent but lower-speed "fender benders" or "pain" incidents (26.63%), whereas nighttime roads are clearer, allowing for higher speeds that result in more severe crashes.

Dusk and Dawn: These transitional periods show a severe injury rate of 2.46%, which is higher than daylight. This is often attributed to the "glare factor" and the biological transition of the human eye adjusting to changing light levels.

8. Weekday-wise Collision Trends

Collision counts were grouped by weekday.



The Friday Peak: There is a significant and visible spike on Friday, reaching the highest point in the dataset (approximately 150,000+ collisions). This is often attributed to the "weekend transition" effect: higher traffic volume as people commute to work while others start weekend travel, combined with potentially higher social activity in the evening.

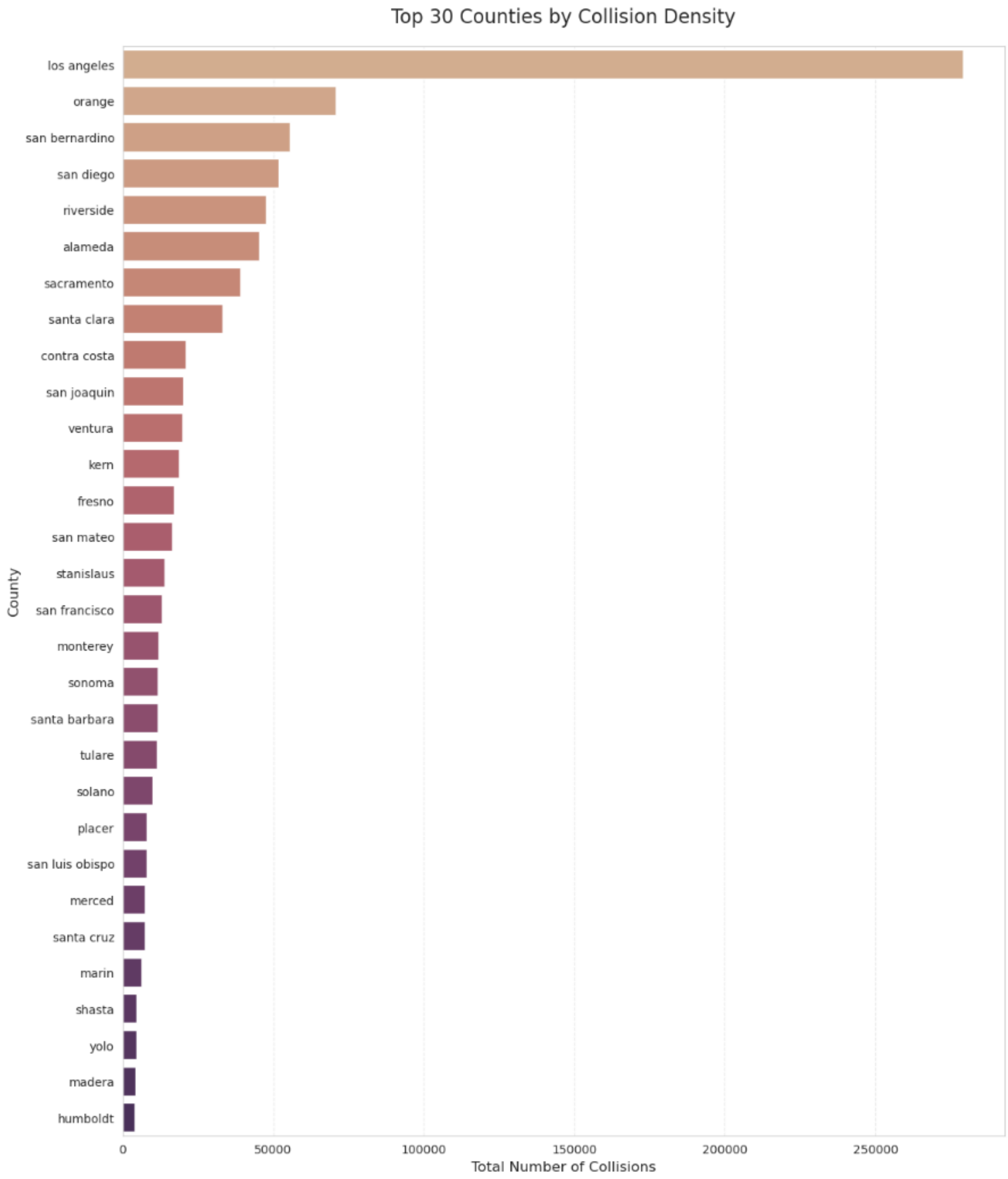
The Mid-Week Plateau: From Tuesday through Thursday, the number of collisions remains remarkably stable, hovering around 135,000. This represents the baseline for standard weekday commuting patterns.

The Sunday Trough: Collisions reach their lowest point on Sunday (dropping toward 100,000). This likely reflects lower overall vehicle miles travelled (VMT), as fewer people are commuting to work or school.

The Monday Reset: Monday shows a slight dip compared to the rest of the workweek. This "Monday effect" is common in traffic data, sometimes attributed to lighter traffic flows or more cautious driving at the start of a new week.

9. County-wise Collision Distribution

Top counties were ranked by collision count.



The Los Angeles Dominance: Los Angeles County is the extreme outlier, with nearly 280,000 collisions. Its count is nearly quadruple that of the second-place county (Orange). This is a direct reflection of L.A.'s massive population, sprawling freeway network, and high vehicle miles travelled (VMT).

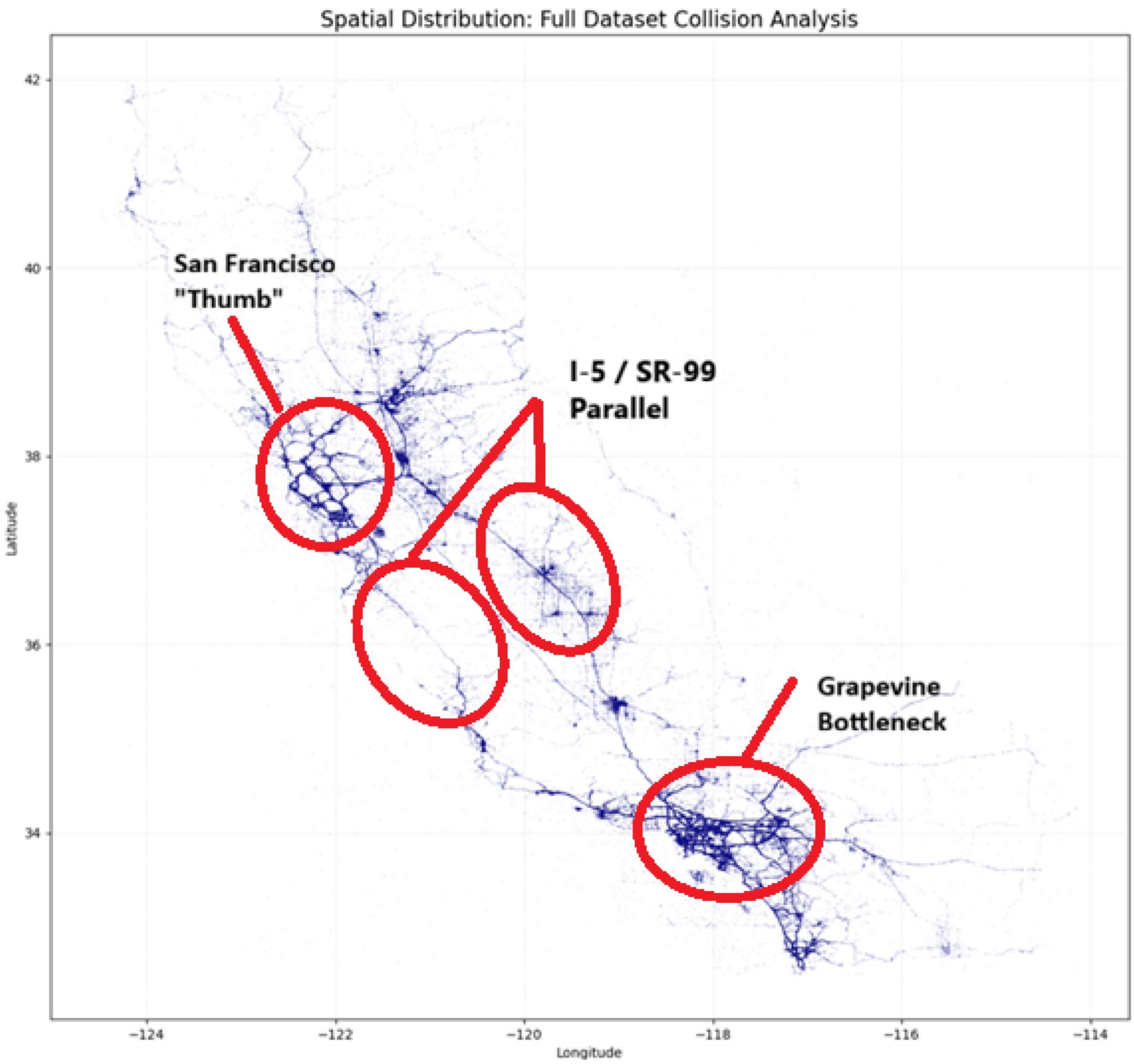
Southern California Concentration: The top five counties (Los Angeles, Orange, San Bernardino, San Diego, and Riverside) are all in Southern California. This cluster represents the highest concentration of traffic risk in the state, likely due to the interconnected nature of the SoCal megalopolis (Southern California's major urban areas).

The "Long Tail" of the Distribution: While the top five counties account for a staggering portion of the total volume, the chart drops off rapidly. By the time you reach Humboldt or Madera (at the bottom of the top 30), the collision counts are a small fraction of L.A.'s total.

Bay Area Presence: Alameda, Santa Clara, and Sacramento represent the primary Northern/Central California hubs, yet even their combined totals struggle to match the sheer volume found in Los Angeles alone.

10. Geographic Scatter Plot Analysis

Latitude and longitude points were plotted on a scatter map.



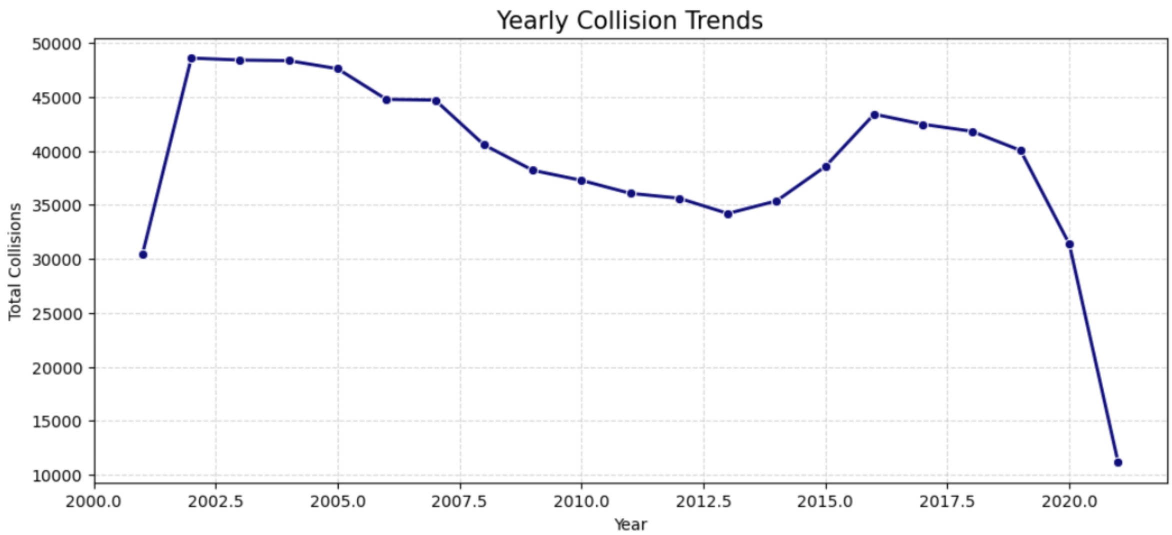
The I-5 / SR-99 Parallel: Between latitudes 35°N and 38°N, we can see two distinct vertical "veins." The western vein is I-5, dominated by long-haul trucking and high-speed transit. The eastern vein is SR-99, which strings together the Central Valley cities (Bakersfield, Fresno, Modesto). The higher density on SR-99 often reflects more frequent intersections and local traffic compared to the more controlled-access I-5.

The "Grapevine" Bottleneck: Around 34.8°N, -118.9°W, there is a noticeable narrowing and intensification where the Central Valley meets the Los Angeles basin. This is the Tejon Pass (The Grapevine), a high-elevation mountain pass where weather and steep grades frequently contribute to the "ghost line" density.

The San Francisco "Thumb": At 37.7°N, -122.4°W, the data points perfectly outline the peninsula. The density here is forced by geography; since there are limited entry/exit points (bridges), the "collision pressure" on those specific coordinates is significantly higher than in the open sprawl of the Inland Empire.

11. Yearly, Monthly, Hourly Trends

Yearly Trends



The Early 2000s Peak (2002–2005): Collisions reached their historical high during this period, plateauing near 48,000–49,000 annually. This correlates with a period of economic growth and high vehicle miles travelled (VMT).

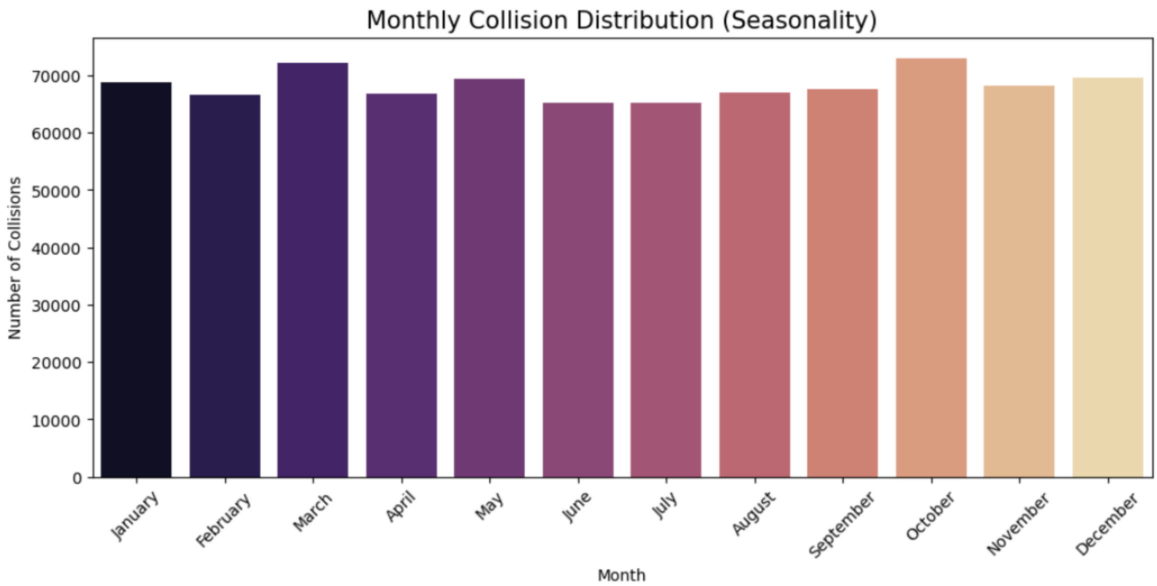
The Great Recession Decline (2006–2013): There is a sustained, dramatic drop in collisions starting around 2007, bottoming out in 2013 at approximately 34,000.

Insight: Economic downturns typically lead to fewer people commuting and less commercial trucking activity, which naturally lowers the "collision exposure."

The Mid-2010s Rebound (2014–2016): As the economy recovered and gas prices fell, collision counts began to climb again, peaking at a secondary high of roughly 43,500 in 2016.

The COVID-19 "Cliff" (2020–2021): The most striking feature of the graph is the vertical drop at the end. Total collisions plummeted from over 30,000 in 2020 to roughly 11,000 in 2021.

Note on Data Integrity: While the 2020 drop is due to lockdowns, the extreme 2021 drop likely indicates an incomplete dataset. Since SWITRS records can take months to finalize, 2021 likely only contains a partial year of data.

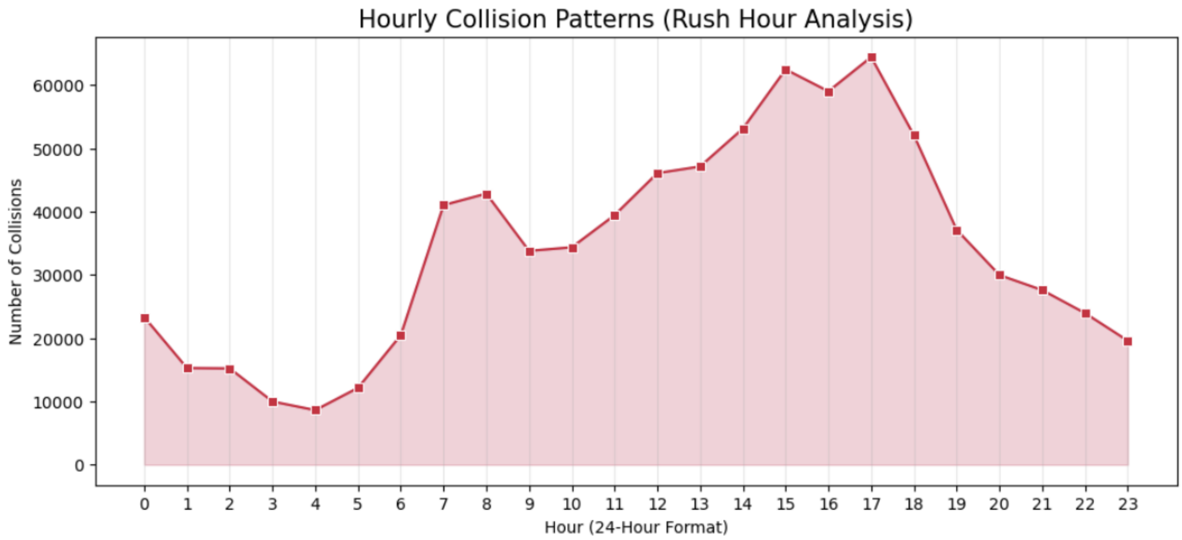


The October Peak: October stands out as the month with the highest number of collisions (exceeding 70,000). This is a well-documented phenomenon in traffic safety, often attributed to the transition into shorter days (earlier sunsets) and the first major rains of the season after a long, dry California summer, which makes roads exceptionally slick due to oil buildup.

The March "Spring Spike": There is a secondary peak in March. This often coincides with Daylight Saving Time transitions (which disrupt sleep patterns) and Spring Break travel.

The Summer Dip (June/July): Interestingly, June and July show lower collision volumes compared to the spring and fall peaks. While people travel more for vacations, the lack of school-related commuting and more consistent, clear daylight conditions may act as a stabilizing factor.

The Year-End Rise: A steady climb is visible from August through October, staying high into December. This reflects the "Holiday Quarter" where increased shopping, social events, and holiday travel, combined with deteriorating weather (rain/fog), create a higher-risk environment.



The Morning Commute (7 AM – 8 AM): There is a sharp, nearly vertical climb starting at 5 AM, peaking around 8 AM. This corresponds with the start of the workday and school morning drop-offs.

The Mid-Day Plateau (10 AM – 2 PM): After the morning rush, volume dips slightly but remains surprisingly high (around 40,000–45,000). This likely represents commercial deliveries and service vehicles.

The Evening Peak (3 PM – 5 PM): This is the most dangerous window of the day. The count hits its absolute maximum at 5 PM (17:00) with over 60,000 collisions.

Why it's higher than the morning: The afternoon rush is typically more sustained. It combines work commuters, school pickups, shoppers, and people heading to social dinner engagements. Additionally, driver fatigue at the end of a workday is a significant contributing factor.

The "Late Night" Trough (3 AM – 4 AM): Collisions reach their lowest point around 4 AM. While accidents during these hours are fewer, they are statistically more likely to involve alcohol or high speeds, which we saw in your "Lighting Conditions" heatmap earlier.

5.4 ETL Querying

1. Load Processed CSV Files to MinIO

   The processed collision dataset was stored in the MinIO object storage bucket and accessed using PySpark. The file was loaded from the specified bucket path into a Spark DataFrame for further analysis. This step ensured that the transformed data was available in a distributed environment for querying and reporting.

   Result: Dataset successfully loaded from s3a://mybucket/collisions_final.parquet.

2. Top 5 Counties by Collisions

   The dataset was grouped by county and the total number of collisions in each county was counted. After sorting in descending order, the top five counties with the highest collision frequency were identified.

Rank	County	Number of Collisions
1	Los Angeles	2,65,550
2	Orange	65,950
3	San Bernardino	47,160
4	San Diego	44,559
5	Alameda	40,645

   Insight: Los Angeles recorded significantly more collisions than any other county, indicating higher traffic density and accident risk.

3. Highest Collision Month

   The collision dates were analysed by extracting the month name and grouping records by month. The month with the maximum number of collisions was then selected.

   Result: October had the highest number of collisions with 72,975 cases.

   Insight: Seasonal factors such as weather changes, traffic congestion, or holiday travel may contribute to increased collisions during October.

4. Most Common Weather Condition

Collision records were grouped by weather condition and counted to determine which weather type was most frequently associated with accidents.

Result: The most common weather condition during collisions was "Clear".

Insight: Most collisions happened during clear weather because normal driving conditions occur more often, leading to higher exposure rather than weather severity alone.

5. Fatal Collision Percentage

Fatal collisions were identified using party records where at least one death occurred. These fatal collision cases were compared against the total number of collisions.

Total Collisions: 819,043

Fatal Collisions: 5,767

Fatal Collision Percentage: 0.70%

Insight: Fatal collisions made up a small percentage of all reported accidents, though they remain highly significant from a public safety perspective.

6. Most Dangerous Time of Day

The hour was extracted from the collision timestamp, and collisions were grouped by hour to determine peak accident times.

Result: The most dangerous time of day was 17:00 (5 PM) with 64,438 collisions.

Insight: Evening rush hour traffic likely contributes to the highest collision frequency due to congestion and commuter movement.

7. Top Road Surface Conditions

Collisions were grouped by road surface type to identify which conditions were most common during accidents.

Rank	Road Surface Condition	Collisions
1	Dry	7,43,485
2	Wet	64,694
3	Unknown	7,651
4	Snowy	2,396
5	Slippery	807

Insight: Most collisions occurred on dry roads because dry conditions are the most common driving environment. Wet roads were the second highest, indicating reduced traction increases risk.

8. Lighting Conditions Analysis

The dataset was analysed based on lighting conditions at the time of collisions. The top three categories with the highest collision counts were identified.

Rank	Lighting Condition	Collisions
1	Daylight	5,49,083
2	Dark with Street Lights	1,85,031
3	Dark with No Street Lights	50,130

Insight: Most collisions occurred during daylight due to higher traffic volume, while nighttime collisions under poor lighting conditions still represent a significant safety concern.

6. Conclusion

This analysis shows that traffic collisions in California are driven less by rare extreme conditions and more by everyday traffic volume, commuting behaviour, infrastructure gaps, and predictable human risk patterns. Most crashes occurred in clear weather, dry roads, and daylight conditions, which means accidents are primarily an exposure problem—not a weather problem. People crash most when roads are busiest, not necessarily when conditions are worst.

Evening rush hour (5 PM) was the single highest-risk time, while Fridays recorded the most collisions. This indicates congestion, fatigue, impatience, and end-of-week traffic pressure are stronger collision drivers than random factors. Traffic enforcement and signal optimization should therefore focus on weekday peak windows rather than distributing resources evenly across the day.

Los Angeles County overwhelmingly led collision volume, followed by other dense urban counties. That means collision prevention should be geographically prioritized. A statewide uniform strategy would waste resources; high-density counties need more targeted interventions such as smarter intersections, faster incident response, lane design changes, and localized enforcement.

Nighttime collisions were fewer in number but more severe, especially in areas without street lighting. This is one of the clearest actionable findings: better lighting infrastructure can reduce severe outcomes faster than many awareness campaigns. Lighting upgrades in dark corridors likely offer measurable ROI.

Young adults (roughly late teens to mid-20s) appeared most frequently among victims, suggesting inexperience, higher exposure, and riskier driving patterns. Driver education, insurance incentives, and targeted safety programs should prioritize this demographic.

From a data perspective, the ETL process successfully transformed messy raw relational records into analysis-ready datasets using Spark and MinIO. Duplicate removal, null handling, type correction, and outlier treatment significantly improved reliability, meaning future dashboards, ML models, and forecasting systems can now be built on a stronger foundation.

Final Business Reality

- If an agency had to act tomorrow, the smartest moves would be:
- Focus enforcement and traffic control during 4 PM–7 PM weekdays.
- Upgrade lighting in dark high-collision corridors.
- Prioritize Los Angeles and other top counties first.
- Build interventions for young drivers.
- Use predictive models to forecast hotspot times and locations.

Bottom Line

The problem is not unpredictable crashes. The problem is repeated collisions in known places, known times, and known conditions. That means it is manageable with targeted, data-led intervention.

6.1 Road Safety Recommendations

- **Target Young Drivers**: Implement targeted safety awareness campaigns for the 18–25 age group, as they represent the primary peak in victim frequency.
- **Enhance Nighttime Infrastructure:** Accelerate the installation and maintenance of street lighting, particularly in high-speed areas, since "dark with no street lights" correlates with the highest severe injury rate (3.88%).
- **Safety Feature Promotion:** Continue promoting advanced vehicle safety features like airbags and crumple zones, which likely contribute to the high volume of non-injury (property damage only) outcomes despite high accident rates.

6.2 Traffic Management Suggestions

- **Rush Hour Mitigation:** Increase traffic monitoring and rapid-response deployments during the 5 PM evening peak, which recorded the highest frequency of accidents (over 64,000 cases).
- **Friday-Specific Protocols:** Develop specific traffic management plans for Fridays to address the significant weekly spike in collisions caused by the "weekend transition" effect.
- **Seasonal Readiness:** Prepare increased maintenance and patrol capacity for October, identified as the highest collision month due to early sunsets and first rains.

6.3 Policy Improvements for Pedestrians & Cyclists

- **Visibility Standards:** Implement stricter lighting and reflective gear policies for nighttime travel, as poor lighting significantly increases the severity of accidents.
- **Urban Speed Management:** Support policies that lower speed limits in high-density urban corridors to convert potentially "severe" injuries into "pain" or "property damage only" incidents.
- **Congestion-Focused Planning:** Prioritize pedestrian and cyclist separation in Southern California hubs where massive traffic volume creates a high-density risk environment.

6.4 High-Risk Zone Identification

- **Southern California Focus:** Los Angeles County is the highest-risk zone, with nearly 280,000 collisions—quadruple the count of the next highest county.
- **The "Grapevine" Bottleneck:** The Tejon Pass (34.8°N, -118.9°W) is a critical high-risk area where elevation and steep grades intensify collision density.
- **Central Valley Arteries:** SR-99 shows higher collision density than I-5 between latitudes 35°N and 38°N, indicating a need for better intersection management in Central Valley cities.

6.5 Environmental Factor Impact

- **Clear Weather Paradox:** While most accidents (678,876) occur in clear weather, this is due to higher exposure; safety efforts should focus on human factors like distraction and speed.
- **Wind and Fog Hazards:** High winds show the highest spike in severe injuries (3.26%), and fog leads to higher rates of "other" and "severe" injuries due to reduced visibility and rear-end collisions.
- **The "Slick Road" Effect:** October's peak is partially attributed to oil buildup on roads being released by the first seasonal rains.

6.6 Predictive Analysis Opportunities

- **Incomplete Data Modelling:** Future models must account for "data integrity" gaps, such as the apparent 2021 decline which likely reflects incomplete reporting rather than a true drop in accidents.
- **Age and Demographic Weighting:** Predictive models can leverage the "age 30" data rounding spike and the "young driver" peak to better estimate risk based on driver profiles.
- **Time-Series Forecasting:** Use the established "Mid-Week Plateau" and "Sunday Trough" to predict daily accident volumes for resource allocation.

6.7 Key Findings & Business Implications

- **Economic Correlation:** Historical trends show collisions decline during economic downturns (2007–2013) and rebound with economic growth, suggesting that traffic risk is a lagging indicator of economic activity.

- **Infrastructure ROI:** The measurable safety benefit of functioning street lights (0.98% reduction in severe injuries compared to unlit areas) provides a clear business case for infrastructure investment.
- **Public Safety Outcome:** While 61% of collisions are property damage only, the 0.70% of fatal collisions remain the highest priority for policy intervention and safety budgeting.